

5.EXAMPLE TEXT W-2

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STEP-1 CREATE DATABASE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', 'Data Import/Restore', 'NCE', 'Startup / Shutdown', 'Server Logs', 'Options File', 'RMANCE', 'Dashboard', 'Performance Reports', and 'Performance Schema Setup'. The right pane shows the 'example.txtw-2' script with the following SQL commands:

```
1 • CREATE DATABASE IF NOT EXISTS joins_db;
2 • USE joins_db;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	15:30:12	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected
2	15:30:12	USE joins_db	0 row(s) affected

STEP-2 CREATE PHYSICIAN TABLE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', 'Data Import/Restore', 'NCE', 'Startup / Shutdown', 'Server Logs', 'Options File', 'RMANCE', 'Dashboard', 'Performance Reports', and 'Performance Schema Setup'. The right pane shows the 'example.txtw-2' script with the following SQL commands:

```
1 • CREATE DATABASE IF NOT EXISTS joins_db;
2 • USE joins_db;
3 • CREATE DATABASE IF NOT EXISTS joins_db;
4 • USE joins_db;
5 •
6 • CREATE TABLE physician (
7 •     employeeid INT PRIMARY KEY,
8 •     name VARCHAR(100),
9 •     position VARCHAR(100),
10 •     ssn VARCHAR(20)
11 • )
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	15:30:12	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected
2	15:30:12	USE joins_db	0 row(s) affected
3	15:34:27	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
4	15:34:27	USE joins_db	0 row(s) affected
5	15:34:27	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100), ssn VARCHAR(20))	0 row(s) affected

STEP-3 INSERT DATA INTO PHYSICIAN

The screenshot shows the SQL Enterprise Manager interface with the following SQL script in the query window:

```
1 CREATE DATABASE IF NOT EXISTS joins_db;
2 USE joins_db;
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K.Vaishalii','Staff Internist','111111111'),
12 (2,'K.Vaishali','Attending Physician','222222222'),
13 (3,'K.Vaishali Turk','Surgical Attending Physician','333333333'),
14 (4,'Percival Cox','Senior Attending Physician','444444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','555555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
17 (7,'John Wen','Surgical Attending Physician','777777777'),
18 (8,'Keith Dudemeister','MD Resident','888888888'),
19 (9,'Molly Clock','Attending Psychiatrist','999999999');
```

The Output pane shows the execution results:

#	Time	Action	Message
8	15:37:35	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
9	15:37:35	USE joins_db	0 row(s) affected
10	15:37:35	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	Error Code: 1050. Table 'physician' already exists
11	15:39:35	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
12	15:39:35	USE joins_db	0 row(s) affected
13	15:39:35	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	Error Code: 1050. Table 'physician' already exists

STEP-4 CREATE THE DEPARTMENT TABLE

The screenshot shows the SQL Enterprise Manager interface with the following SQL script in the query window:

```
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
9
10 INSERT INTO physician VALUES
11 (1,'K.Vaishalii','Staff Internist','111111111'),
12 (2,'K.Vaishali','Attending Physician','222222222'),
13 (3,'K.Vaishali Turk','Surgical Attending Physician','333333333'),
14 (4,'Percival Cox','Senior Attending Physician','444444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','555555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
17 (7,'John Wen','Surgical Attending Physician','777777777'),
18 (8,'Keith Dudemeister','MD Resident','888888888'),
19 (9,'Molly Clock','Attending Psychiatrist','999999999');
20 CREATE TABLE department (
21     departmentid INT PRIMARY KEY,
22     name VARCHAR(100),
23     head INT
24 );
```

The Output pane shows the execution results:

#	Time	Action	Message
11	15:39:35	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
12	15:39:35	USE joins_db	0 row(s) affected
13	15:39:35	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	Error Code: 1050. Table 'physician' already exists
14	15:43:54	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db'; database exists
15	15:43:54	USE joins_db	0 row(s) affected
16	15:43:54	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	Error Code: 1050. Table 'physician' already exists

STEP-5 INSERT DATA INTO DEPARTMENT

Automatic context help is disabled. Use the toolbar to manually
caret position or to toggle automatic help

```
8 }
9
10 INSERT INTO physician VALUES
11 (1,'K.Vaishali','Staff Internist','11111111'),
12 (2,'K.Vaishali','Attending Physician','22222222'),
13 (3,'K.Vaishali Turk','Surgical Attending Physician','33333333'),
14 (4,'Percival Cox','Senior Attending Physician','44444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','55555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','66666666'),
17 (7,'John Wen','Surgical Attending Physician','77777777'),
18 (8,'Keith Dudemeister','MD Resident','88888888'),
19 (9,'Molly Clock','Attending Psychiatrist','99999999');
20
21 CREATE TABLE department (
22     departmentid INT PRIMARY KEY,
23     name VARCHAR(100),
24     head INT
25 );
26
27 INSERT INTO department VALUES
28 (1,'General Medicine',4),
29 (2,'Surgery',7),
30 (3,'Psychiatry',9);
```

Output:

#	Time	Action	Message
12	15:39:35	USE joins_db	0 row(s) affected
13	15:39:35	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100))	Error Code: 1050. Table 'physician' already exists
14	15:43:54	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'joins_db': database exists
15	15:43:54	USE joins_db	0 row(s) affected
16	15:43:54	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100))	Error Code: 1050. Table 'physician' already exists
17	15:45:21	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	Error Code: 1146. Table 'joins_db.department' doesn't exist

STEP-6 INNER JOIN

Automatic context help is disabled. Use the toolbar to manually
caret position or to toggle automatic help

```
33 (2,'Surgery',7),
34 (3,'Psychiatry',9);
35
36 SELECT d.name AS Department,
37        p.name AS Head_Physician
38 FROM department d
39 JOIN physician p
40 ON d.head = p.employeeid;
41
```

Result Grid:

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

Output:

#	Time	Action	Message
27	15:51:51	DROP TABLE IF EXISTS physician	0 row(s) affected
28	15:51:51	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100))	0 row(s) affected
29	15:51:51	INSERT INTO physician VALUES (1,'K.Vaishali','Staff Internist','11111111'), (2,'K.Vaishali','Attending Physician','22222222'), (3,'K.Vaishali Turk','Surgical Attending Physician','33333333'), (4,'Percival Cox','Senior Attending Physician','44444444'), (5,'Bob Kelso','Head Chief of Medicine','55555555'), (6,'Todd Quinlan','Surgical Attending Physician','66666666'), (7,'John Wen','Surgical Attending Physician','77777777'), (8,'Keith Dudemeister','MD Resident','88888888'), (9,'Molly Clock','Attending Psychiatrist','99999999');	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
30	15:51:51	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
31	15:51:51	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
32	15:51:51	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid;	3 row(s) returned

STEP-7 WHERE CLAUSE

The screenshot shows a database management tool interface. The main window displays a SQL query in a text editor:

```
35
36 • SELECT d.name AS Department,
37       p.name AS Head_Physician
38 FROM department d
39 JOIN physician p
40 ON d.head = p.employeeid;
41 • SELECT *
42 FROM physician
43 WHERE position='Surgical Attending Physician';
```

Below the query editor, the 'Result Grid' shows the results of the query:

	employeeid	name	position	ssn
▶	3	K.Vaishali Turk	Surgical Attending Physician	333333333
	6	Todd Quinlan	Surgical Attending Physician	666666666
	7	John Wen	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

On the right side, a message states: "Automatic context help is disabled. Use the caret position or to l".

At the bottom, the 'Output' pane shows the execution log:

#	Time	Action	Message
✓ 28	15:51:51	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHA...	0 row(s) affected
✓ 29	15:51:51	INSERT INTO physician VALUES (1,'K.Vaishali','Staff Internist','111111111'), (2,'K.Vaishali','Attending Physicia...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 30	15:51:51	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 31	15:51:51	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 32	15:51:51	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 33	15:52:55	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned

STEP-8 WHERE + LIKE

K.Vaishali Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2

SQL Additions

Automatic context help is disabled. Use the caret position or to l

38 FROM department d
39 JOIN physician p
40 ON d.head = p.employeeid;
41 • SELECT *
42 FROM physician
43 WHERE position='Surgical Attending Physician';
44 • SELECT *
45 FROM physician
46 WHERE name LIKE 'K.Vaishali%';

Result Grid

employeeid	name	position	ssn
1	K.Vaishali	Staff Internist	111111111
2	K.Vaishali	Attending Physician	222222222
3	K.Vaishali Turk	Surgical Attending Physician	333333333
•	NULL	NULL	NULL

Result 3 physician 4 physician 5 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 40	15:55:17	INSERT INTO physician VALUES (1,'K.Vaishali','Staff Internist','111111111'), (2,'K.Vaishali','Attending Physicia...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 41	15:55:17	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 42	15:55:17	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 43	15:55:17	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 44	15:55:17	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned
✓ 45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned

STEP-9 COUNT()

SQL workbench

K.Vaishali Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2

SQL Additions

Automatic context help is disabled. Use the caret position or to l

40 ON d.head = p.employeeid;
41 • SELECT *
42 FROM physician
43 WHERE position='Surgical Attending Physician';
44 • SELECT *
45 FROM physician
46 WHERE name LIKE 'K.Vaishali%';
47 • SELECT COUNT(*) AS Total_Physicians
48 FROM physician;

Result Grid

Total_Physicians
9

Result 6 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 41	15:55:17	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓ 42	15:55:17	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 43	15:55:17	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓ 44	15:55:17	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned
✓ 45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned
✓ 46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned

STEP-10 COUNT DISTINCT

KVaishai Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.btw-2

SQL Additions

Automatic context help is disabled. Use the caret position or to

```

42 FROM physician
43 WHERE position='Surgical Attending Physician';
44 • SELECT *
45 FROM physician
46 WHERE name LIKE 'K.Vaishali%';
47 • SELECT COUNT(*) AS Total_Physicians
48 FROM physician;
49 • SELECT COUNT(DISTINCT position)
50 FROM physician;

```

Result Grid

	COUNT(DISTINCT position)
7	7

Result 7

Output

Action Output

#	Time	Action	Message
42	15:55:17	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',3)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
43	15:55:17	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
44	15:55:17	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned
45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned
46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned
47	15:57:35	SELECT COUNT(DISTINCT position) FROM physician	1 row(s) returned

STEP-11 GROUP BY

KVaishai Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.btw-2

SQL Additions

Automatic context help is disabled. Use the caret position or to

```

46 WHERE name LIKE 'K.Vaishali%';
47 • SELECT COUNT(*) AS Total_Physicians
48 FROM physician;
49 • SELECT COUNT(DISTINCT position)
50 FROM physician;
51 • SELECT position,
52 COUNT(*) AS Total_Employees
53 FROM physician
54 GROUP BY position;

```

Result Grid

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 8

Output

Action Output

#	Time	Action	Message
43	15:55:17	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
44	15:55:17	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned
45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned
46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned
47	15:57:35	SELECT COUNT(DISTINCT position) FROM physician	1 row(s) returned
48	15:58:36	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position	7 row(s) returned

STEP-12 GROUP BY + HAVING

KVaishali Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2*

SQLAdditions

Automatic context he

51 SELECT position,
52 COUNT(*) AS Total_Employees
53 FROM physician
54 GROUP BY position;
55 SELECT position,
56 COUNT(*) AS Total_Employees
57 FROM physician
58 GROUP BY position
59 HAVING COUNT(*) > 1;

Result Grid Filter Rows: Total_Employees

position	Total_Employees
Surgical Attending Physician	3

Result 9 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
44	15:55:17	SELECT * FROM physician WHERE position='Surgical Attending Physician'	3 row(s) returned
45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned
46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned
47	15:57:35	SELECT COUNT(DISTINCT position) FROM physician	1 row(s) returned
48	15:58:36	SELECT position, COUNT(*) FROM physician GROUP BY position	7 row(s) returned
49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

object selected

STEP-13 ORDER BY

KVaishali Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2*

SQLAdditions

Automatic context h

54 GROUP BY position;
55 SELECT position,
56 COUNT(*) AS Total_Employees
57 FROM physician
58 GROUP BY position
59 HAVING COUNT(*) > 1;
60 SELECT *
61 FROM physician
62 ORDER BY name ASC;

Result Grid Filter Rows: physician 10 x Apply Context Help Snippets

employeeid	name	position	ssn
5	Bob Kelso	Head Chief of Medicine	55555555
7	John Wen	Surgical Attending Physician	77777777
2	K.Vaishali	Attending Physician	22222222
3	K.Vaishali Turk	Surgical Attending Physician	33333333
1	K.Vaishali	Staff Internist	11111111
8	Keith Dudemeister	MD Resident	88888888
9	Molly Clock	Attending Psychiatrist	99999999
4	Perceval Cox	Senior Attending Physician	44444444
6	Todd Quinlan	Surgical Attending Physician	66666666

Output

Action Output

#	Time	Action	Message
45	15:55:17	SELECT * FROM physician WHERE name LIKE 'K.Vaishali%'	3 row(s) returned
46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned
47	15:57:35	SELECT COUNT(DISTINCT position) FROM physician	1 row(s) returned
48	15:58:36	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position	7 row(s) returned
49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned

object selected

STEP-14 ORDER BY DESC

KVaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.btw-2*

SQLAdditions

Automatic context h

```

57 FROM physician
58 GROUP BY position
59 HAVING COUNT(*) > 1;
60 • SELECT *
61 FROM physician
62 ORDER BY name ASC;
63 • SELECT *
64 FROM physician
65 ORDER BY employeeid DESC;

```

Result Grid

employeeid	name	position	ssn
9	Molly Clock	Attending Psychiatrist	999999999
8	Keith Dudemeister	MD Resident	888888888
7	John Wen	Surgical Attending Physician	777777777
6	Todd Quinlan	Surgical Attending Physician	666666666
5	Bob Kelso	Head Chief of Medicine	555555555
4	Percival Cox	Senior Attending Physician	444444444
3	K.Vaishali Turk	Surgical Attending Physician	333333333
2	K.Vaishali	Attending Physician	222222222
1	K.Vaishali	Staff Internist	111111111

physician 11

Output

Action Output

#	Time	Action	Message
✓ 46	15:56:33	SELECT COUNT(*) AS Total_Physicians FROM physician	1 row(s) returned
✓ 47	15:57:35	SELECT COUNT(DISTINCT position) FROM physician	1 row(s) returned
✓ 48	15:58:36	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position	7 row(s) returned
✓ 49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned
✓ 51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned

Info Session

STEP-15 WHERE + IN

KVaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.btw-2*

SQLAdditions

Automatic context h

```

64 FROM physician
65 ORDER BY employeeid DESC;
66 • SELECT *
67 FROM physician
68 WHERE employeeid IN
69 (
70 SELECT head
71 FROM department
72 );

```

Result Grid

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	444444444
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

physician 12

Output

Action Output

#	Time	Action	Message
✓ 47	15:57:31	Select output pane	
✓ 48	15:58:36	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position	7 row(s) returned
✓ 49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned
✓ 51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned
✓ 52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned

Info Session

STEP-16 SUBQUERY (NOT IN)

K.Vaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.txtw-2" x

SQLAdditions

Automatic context help

```

65 ORDER BY employeeid DESC;
66 SELECT *
67 FROM physician
68 WHERE employeeid IN
69 (
70     SELECT head
71     FROM department
72 );
73 SELECT *

```

Result Grid

employeeid	name	position	ssn
1	K. Vaishali	Staff Internist	111111111
2	K. Vaishali	Attending Physician	222222222
3	K. Vaishali Turk	Surgical Attending Physician	333333333
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
8	Keith Dudemeister	MD Resident	888888888

physician 13 x

Output

Action Output

#	Time	Action	Message
48	15:58:36	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position	7 row(s) returned
49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned
51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned
52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned

Info Session

STEP-17 GROUP BY + ORDER BY

K.Vaishali Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2" x

SQLAdditions

Automatic context help

```

76 (
77     SELECT head
78     FROM department
79 );
80 SELECT position,
81     COUNT(*) AS Total_Employees
82 FROM physician
83 GROUP BY position
84 ORDER BY Total_Employees DESC;

```

Result Grid

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 14 x

Output

Action Output

#	Time	Action	Message
49	15:59:42	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned
51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned
52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned
54	16:06:25	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned

Info Session

STEP-18 WHERE + LIKE

KVaishai Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2'

SQLAdditions

Automatic context help

79);

80 SELECT position,

81 COUNT(*) AS Total_Employees

82 FROM physician

83 GROUP BY position

84 ORDER BY Total_Employees DESC;

85 SELECT *

86 FROM physician

87 WHERE position LIKE '%Attending%';

Result Grid

	employeeid	name	position	ssn
2	K.Vaishali	Attending Physician	222222222	
3	K.Vaishali Turk	Surgical Attending Physician	333333333	
4	Percival Cox	Senior Attending Physician	444444444	
6	Todd Quinlan	Surgical Attending Physician	666666666	
7	John Wen	Surgical Attending Physician	777777777	
9	Molly Clock	chiatrist	999999999	

physician 15 x

Output

#	Time	Action	Message
50	16:01:21	SELECT * FROM physician ORDER BY name ASC	9 row(s) returned
51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned
52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned
54	16:06:25	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
55	16:07:56	SELECT * FROM physician WHERE position LIKE '%Attending%'	6 row(s) returned

Info Session

STEP-19 JOIN + WHERE

KVaishai Local instanceEXT...

Edit View Query Database Server Tools Scripting Help

example.txtw-2'

SQLAdditions

Automatic context help

84 ORDER BY Total_Employees DESC;

85 SELECT *

86 FROM physician

87 WHERE position LIKE '%Attending%';

88 SELECT p.name

89 FROM physician p

90 JOIN department d

91 ON p.employeeid=d.head

92 WHERE d.name='Surgery';

Result Grid

	name
John Wen	

Result 16 x

Output

#	Time	Action	Message
51	16:02:36	SELECT * FROM physician ORDER BY employeeid DESC	9 row(s) returned
52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned
54	16:06:25	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
55	16:07:56	SELECT * FROM physician WHERE position LIKE '%Attending%'	6 row(s) returned
56	16:09:23	SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery'	1 row(s) returned

Info Session

STEP-20 JOIN + ORDER BY

KVaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.txtw-2

SQLAdditions

Automatic context h

```

90 JOIN department d
91 ON p.employeeid=d.head
92 WHERE d.name='Surgery';
93 SELECT d.name,
94        p.name
95 FROM department d
96 JOIN physician p
97 ON d.head=p.employeeid
98 ORDER BY d.name;

```

Result Grid

name	name
General Medicin	General Medicine
Psychiatry	Molly Clock
Surgery	John Wen

Result 17

Output

#	Time	Action	Message
52	16:03:52	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department)	3 row(s) returned
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned
54	16:06:25	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
55	16:07:56	SELECT * FROM physician WHERE position LIKE '%Attending%'	6 row(s) returned
56	16:09:23	SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery'	1 row(s) returned
57	16:10:33	SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d...	3 row(s) returned

STEP-21 WHERE + NOT LIKE

KVaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.txtw-2

SQLAdditions

Automatic context h

```

93 SELECT d.name,
94        p.name
95 FROM department d
96 JOIN physician p
97 ON d.head=p.employeeid
98 ORDER BY d.name;
99 SELECT *
100 FROM physician
101 WHERE position NOT LIKE '%Surgical%';

```

Result Grid

employeeid	name	position	ssn
1	K.Vaishali	Staff Internist	111111111
2	K.Vaishali	Attending Physician	222222222
4	Perdval Cox	Senior Attending Physician	444444444
5	Bob Kelso	Head Chief of Medicine	555555555
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

physician 18

Output

#	Time	Action	Message
53	16:05:14	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department)	6 row(s) returned
54	16:06:25	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
55	16:07:56	SELECT * FROM physician WHERE position LIKE '%Attending%'	6 row(s) returned
56	16:09:23	SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery'	1 row(s) returned
57	16:10:33	SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d...	3 row(s) returned
58	16:12:05	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%'	6 row(s) returned

STEP-22 COUNT + HAVING

gator

example.txtw-2"

SQLAdditions

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K.Vaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.txtw-2*

Don't Limit

SQLAdditions

Automatic context h

```

106 HAVING COUNT(*) >= 2;
107 SELECT *
108 FROM department d
109 WHERE EXISTS
110 (
111 SELECT 1
112 FROM physician p
113 WHERE p.employeeid=d.head
114 );

```

Result Grid

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	Psychiatry

department 20

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message
55	16:07:56	SELECT * FROM physician WHERE position LIKE '%Attending%'	6 row(s) returned
56	16:09:23	SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery'	1 row(s) returned
57	16:10:33	SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d...	3 row(s) returned
58	16:12:05	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%'	6 row(s) returned
59	16:13:44	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2	1 row(s) returned
60	16:14:59	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid...	3 row(s) returned

Info Session

STEP-24 MAX()

K.Vaishali Local instanceEXT...

File Edit View Query Database Server Tools Scripting Help

example.txtw-2*

Don't Limit

SQLAdditions

Automatic context h

```

108 FROM department d
109 WHERE EXISTS
110 (
111 SELECT 1
112 FROM physician p
113 WHERE p.employeeid=d.head
114 );
115 SELECT MAX(employeeid)
116 FROM physician;

```

Result Grid

MAX(employeeid)
9

Result 21

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
56	16:09:23	SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery'	1 row(s) returned
57	16:10:33	SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d...	3 row(s) returned
58	16:12:05	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%'	6 row(s) returned
59	16:13:44	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2	1 row(s) returned
60	16:14:59	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid...	3 row(s) returned
61	16:15:50	SELECT MAX(employeeid) FROM physician	1 row(s) returned

Info Session

STEP-25 MIN()

The screenshot displays the Oracle SQL Developer environment. The main window shows a query window with the following SQL code:

```

110 (
111     SELECT 1
112     FROM physician p
113     WHERE p.employeeid=d.head
114 );
115 • SELECT MAX(employeeid)
116     FROM physician;
117 • SELECT MIN(employeeid)
118     FROM physician;

```

The query is executed, and the result grid shows the following data:

MIN(employeeid)
1

The bottom window shows the log of SQL statements executed during the session:

#	Time	Action	Message
57	16:10:33	SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d...	3 row(s) returned
58	16:12:05	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%'	6 row(s) returned
59	16:13:44	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2	1 row(s) returned
60	16:14:59	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid...	3 row(s) returned
61	16:15:50	SELECT MAX(employeeid) FROM physician	1 row(s) returned
62	16:17:05	SELECT MIN(employeeid) FROM physician	1 row(s) returned